

Microbicidal activity of TiO₂ nanoparticles synthesised by sol–gel method

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Abstract: In this study, the authors investigated antimicrobial activity of TiO₂ nanoparticles (NPs) synthesised by sol–gel method. As synthesised TiO₂ NPs were characterised by X-ray diffraction, scanning electron microscopy and ultraviolet-visible absorption spectroscopy. The antimicrobial activity of calcined TiO₂ nanoparticle samples was examined in day light on Gram positive bacteria (*Staphylococcus aureus*, *Streptococcus pneumoniae* and *Bacillus subtilis*), Gram negative bacteria (*Proteus vulgaris*, *Pseudomonas aeruginosa* and *Escherichia coli*) and fungal test pathogen *Candida albicans*. The synthesised TiO₂ NPs were found to be effective in visible light against *Streptococcus pneumoniae*, *Staphylococcus aureus*, *Proteus vulgaris*, *Pseudomonas aeruginosa* and *Candida albicans*.

1 Introduction

The emergence of infectious diseases poses a serious threat to public health worldwide, especially with the emergence of antibiotic-resistant bacterial strains. Survival of these microbes on environment and medical devices causes nosocomial disease transmission. Over the years, antibiotics have been used to control infections resulting from community and hospital environments [1–3]. Such infections are frequently connected with resistant organisms, including methicillin-resistant *Staphylococcus aureus* (MRSA), penicillin-resistant *Streptococcus pneumoniae*, multidrug-resistant *Acinetobacter baumannii* and multidrug-resistant *Pseudomonas aeruginosa* [4, 5].

Advances in the field of nanobiotechnology lead to the development of new antibacterial agents. Considering the unique properties of nanoparticles (NPs), nano-sized organic and inorganic particles are being generated for ultimate use in medical practices, such as metal oxides of zinc, copper, and iron in biomedical research and antimicrobial activity [6–8]. Antimicrobial activity of NPs has largely been studied with human pathogenic bacteria such as *Escherichia coli* (*E. coli*) [9] and *Staphylococcus aureus* [10]. Bactericidal activity of NPs in part depends on size, stability and concentration in the growth medium. While growing in medium amended with NPs, the bacterial population growth can be inhibited by specific nanoparticle interactions [11]. In general, bacterial cell size is in the micrometre range, while its outer cellular membranes have pores in the nanometre range. NPs have a unique ability of crossing the cell membrane because their sizes are smaller than bacterial pores.

Nanocrystalline Titanium dioxide (TiO₂) is a well-known semi-conductor with photocatalytic activities, having great potential as a self-cleaning and self-disinfecting material in many applications. TiO₂ has been used broadly for killing different groups of microorganisms including bacteria, fungi and viruses, because it has high photoreactivity, broad-spectrum antibiosis and chemical stability [12–19]. TiO₂ NPs decompose organic compounds by the formation and constant release of hydroxyl radicals and superoxide ions when exposed to non-lethal ultraviolet (UV) light, which is highly efficient in inhibiting the growth of MRSA [20]. This potent oxidising power of TiO₂ NPs can be used against bacteria and other organic substances [21–23].

In comparison with published reports on physical, chemical and biological properties, very limited information is available on the broad spectrum of antimicrobial properties of TiO₂ NPs in visible light. The present study is designed to evaluate the antimicrobial activity of TiO₂ NPs synthesised by sol–gel method. As prepared TiO₂ NPs samples calcined at three different temperatures, 400, 600 and 800°C, are used against the microbes such as Gram positive bacteria: *Staphylococcus aureus*, *Streptococcus pneumoniae*, *Bacillus subtilis*; Gram negative bacteria: *Proteus vulgaris*, *Pseudomonas aeruginosa*, *E. coli* and the fungal test pathogen: *Candida albicans*.

2 Materials and methods

Titanium (IV)-n-butoxide (Ti(OBu)₄) (98%, Alfa Aesar), isopropyl alcohol (99%, Fisher Scientific) and HNO₃ (Merck) were used without further purification for the synthesis of nanocrystalline TiO₂. Distilled water was used in all synthesis procedures. Synthesis procedures for the preparation of TiO₂ NPs used in this work were described elsewhere [24]. The precursor solution was a mixture of 5 cc titanium (IV)-n-butoxide (Ti(OBu)₄) and 10 cc isopropyl alcohol. This solution was then added drop wise to a pH 2 solution of HNO₃. The hydrolysis was performed at room temperature while stirring for 3 h using magnetic stirrer. After complete hydrolysis of Ti(OBu)₄, the solution was refluxed at 70°C for 20 h and a sol was formed. Finally, the formed sol was dried at 60°C for 36 h in a hot air oven to obtain TiO₂ powders. The obtained powder samples were calcined for 3 h in a muffle furnace at 400, 600 and 800°C in an air atmosphere and are denoted by S1, S2 and S3 respectively.

2.1 Characterisation of TiO₂ NPs

X-ray diffraction (XRD) techniques used in this work were described by our previous study [24]. XRD analysis was carried out in reflection mode with CuK_α radiation ($\lambda = 1.5406 \text{ \AA}$) using Bruker D8 advance X-ray diffractometer (step size = 0.020°, step time = 31.2 s and 40 kV, 35 mA) with CuK_α radiation in 2 θ range from 20 to 70°. The microstructures of the TiO₂ samples S1, S2 and S3

were taken using a JOEL Model JSM-6390 LV scanning electron microscope (SEM). UV–Vis absorption spectra of the samples were recorded in wavelength range 200–800 nm with a Shimadzu UV–Vis spectrophotometer, UV 2600 model. For measurements, NPs were pressed into a thick pellet and placed at entrance port of integrating sphere using a sample holder. Calibration of absorbance scale was done using standard reference materials.

2.2 Source of microorganisms

Microbial strains obtained from the microbial type culture collection (MTCC), Chandigarh. Stock cultures of different Gram-negative: *E. coli* (MTCC No: 443), *Proteus vulgaris* (MTCC No: 1771) and *Pseudomonas aeruginosa* (MTCC No: 424); Gram-positive: *Bacillus subtilis* (MTCC No: 10619), *Streptococcus pneumonia* (MTCC No: 2672) and *Staphylococcus aureus* (MTCC No: 96)

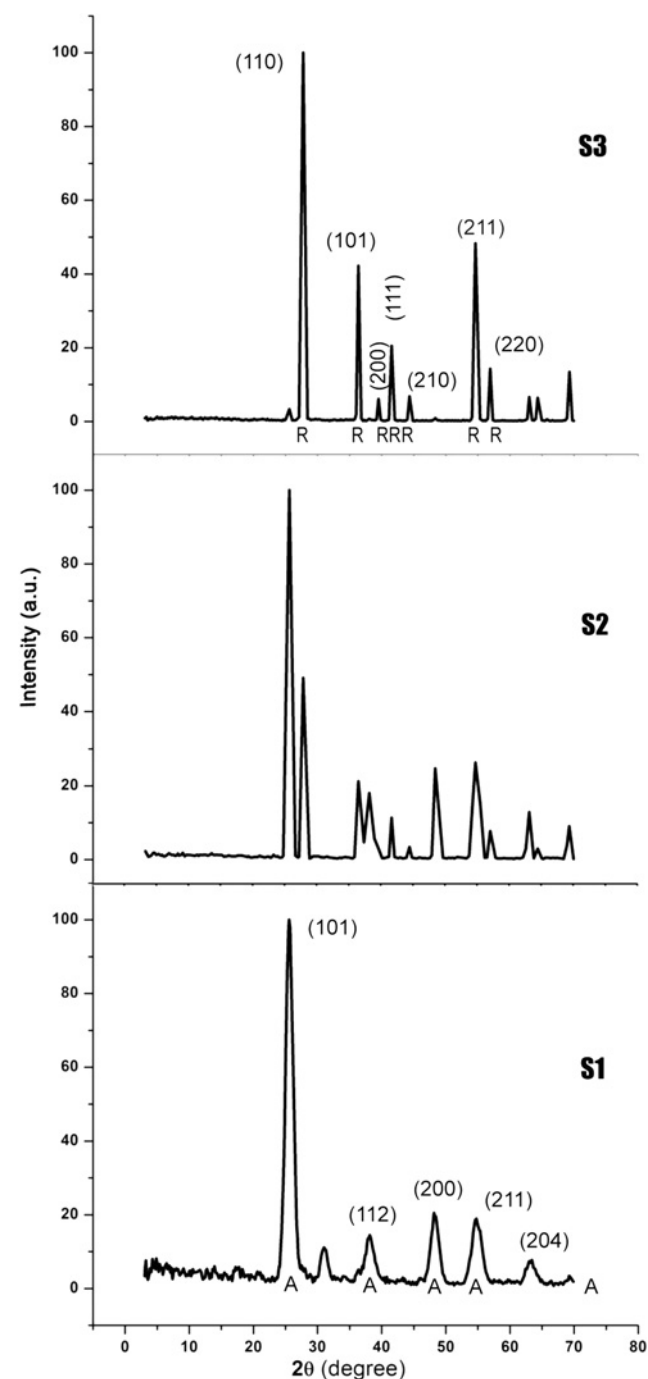


Fig. 1 XRD spectra of TiO_2 samples S1, S2 and S3

bacteria and fungus *Candida albicans* (MTCC No: 227) were sub-cultured and maintained in nutrient broth/Brain heart Infusion broth and Todd's Hewitt Broth.

2.3 Preparation of TiO_2 NPs suspension

150, 200 and 250 microgram (μg) of TiO_2 NPs was added to 1 ml of sterile MQ water and shaken vigorously. The suspending solution was treated by ultrasound (100 W, 40 kHz) for 30 min, autoclaved at 121°C for 20 min and then cooled down to room temperature. The sonication was used to break down the agglomerates and aggregation of NPs to make their dispersions.

2.4 Screening of antibiotics

The antibiotic susceptibility pattern was determined by modified Kirby Bauer disc diffusion method [25] against the following antibiotics (drug concentration in μl): Streptomycin (30), ampicillin (10), Erythromycin (30), Tetracycline (30) for bacterial culture and amphotericin B and chloramphenicol (30) for fungal culture. The Mueller–Hinton agar (MHA) plate supplemented with Gram positive (*Streptococcus pneumoniae*, *Staphylococcus aureus*, *Bacillus subtilis*), Gram negative (*Proteus vulgaris*, *Pseudomonas aeruginosa*) and fungus (*Candida albicans*) by a sterile cotton swab and the antibiotics discs were laid on the surface to find antibiotic resistance pattern. It was found that Tetracycline was more sensitive against gram positive and gram negative bacteria. For fungus, chloramphenicol showed more sensitivity than amphotericin. Thus, tetracycline (30 $\mu\text{l}/\text{disc}$) and chloramphenicol (30 $\mu\text{l}/\text{disc}$) were selected as the standards for bacteria and fungus respectively, in antibiotic sensitivity testing.

2.5 Antimicrobial assay

Antimicrobial activities of the synthesised metal oxide (TiO_2) NPs were performed against Gram positive bacteria (*Streptococcus pneumoniae*, *Staphylococcus aureus*, *Bacillus subtilis*), Gram negative bacteria (*Proteus vulgaris*, *Pseudomonas aeruginosa*, *E. coli*) and Fungus (*Candida albicans*). The Kirby–Bauer disc diffusion method was used for the antimicrobial assay [25]. Using a sterile cotton swab lawn, cultures of the test organisms were made on nutrient agar plates under aseptic conditions. Plates were left standing for 10 min to let the culture get absorbed. Then, 8 mm wells were punched into the MHA plates for testing the antimicrobial activity of TiO_2 nanoparticle samples S1, S2 and S3. Wells were sealed with one drop of molten agar (0.8% agar) to prevent leakage of nanomaterials from the bottom of the wells. Using a micropipette, 30 μl of each sample at different concentrations (150, 200 and 250 $\mu\text{g}/\text{ml}$) of nanoparticle suspension was poured into each wells on three sets of petriplate taken for S1, S2 and S3. The plates are incubated at $37 \pm 1^\circ\text{C}$ and $27 \pm 1^\circ\text{C}$ respectively for bacteria and fungus for a period of 12–24 h both in dark and visible light. After incubation the zone diameter was measured.

3 Results and discussion

The XRD patterns of calcined TiO_2 samples are shown in Fig. 1. The XRD pattern of the sample calcinated at 400°C (S1) reveals the presence of anatase phase. Rutile peaks started to appear and a mixture of both the anatase and rutile phases of TiO_2 existed at 600°C (S2). As calcinations temperature was increased further, a phase transition from anatase to rutile was observed, and the transition to rutile phase is almost completed at 800°C (S3). The average crystallite size of TiO_2 NPs were estimated using Scherrer's equation, $t = 0.9\lambda/\beta\cos\theta$, where λ is the X-ray wavelength, β the full width at half maximum (FWHM) of the peak and θ the Bragg's angle. The size of the anatase crystallites (S1) increases from 6 to 25.8 nm when the calcination temperature is raised to 600°C (S2), and increases to 33 nm as temperature is

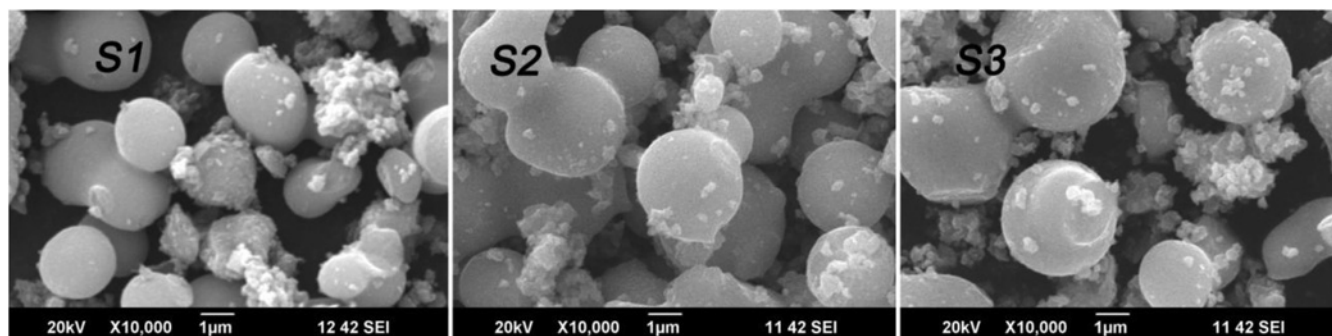


Fig. 2 SEM images of TiO₂ samples S1, S2 and S3

further raised to 800°C (S3, rutile). The XRD spectra shows that the diffraction peaks become intense and their FWHM gradually decreases with increasing calcinations temperature, verifying the increase in particle size. At higher calcinations temperatures, the formed crystallites are larger in size, which can be attributed to the thermally promoted crystallite growth [24].

All the peaks of anatase TiO₂ sample resemble to a tetragonal structure (space group: $I4_1/amd$) containing twelve atoms per unit cell with lattice parameters $a = 3.776 \text{ \AA}$ and $c = 9.486 \text{ \AA}$, which is in agreement with the JCPDS file No. 73-1764 for TiO₂. All the XRD peaks of rutile TiO₂ sample correspond to a tetragonal structure (space group: $P4_2/mnm$) with lattice parameters $a =$

$4.582 \text{ \AA}$ and $c = 2.953 \text{ \AA}$, which is in agreement with the JCPDS file No. 78-1510 for TiO₂.

Fig. 2 shows SEM images of the samples S1, S2 and S3 respectively. The micrograph shows polycrystalline texture of the material with a large number of interfaces. It shows that all samples have almost spherical shaped grains with some voids are distributed almost uniformly over the entire surface of samples. Moreover, grains of unequal sizes are distributed throughout the samples. Increase of particle size due to calcination is also evident from SEM images of samples.

Fig. 3 shows UV-Vis absorbance spectra of TiO₂ nanoparticle samples. It is observed that absorbance spectrum of anatase TiO₂ is blue shifted due to quantum size effect. Direct band gap (E_g) of samples is determined by fitting absorption data to direct transition equation, $ah\nu = E_d (h\nu - E_g)^{1/2}$, where α is optical absorption coefficient, $h\nu$ photon energy, E_g direct band gap and E_d a constant. Band gap of TiO₂ samples have been measured by plotting $(ah\nu)^2$ as a function of photon energy and extrapolating linear portion of the curve to absorption equal to zero. The band gap values obtained are 2.87, 2.76 and 2.67 eV for samples S1, S2 and S3 respectively. These values are lower than their bulk and literature values [26, 27]. This lowering of band gap may be due to chemical defects or vacancies present in crystal generating new energy levels. The small band gap values of TiO₂ NP samples are useful for higher photocatalytic as well as antimicrobial activities because their energy gap values are comparable with energy of visible or UV light photons. It is also found that band gap decreases with increase in calcination due to particle size variation.

The TiO₂ NPs synthesised at three different temperatures were used for checking the antibiotic sensitivity at three different concentrations (150, 200 and 250 µg/ml). No significant antimicrobial activity was detected for samples kept in the dark for 12–24 h. Table 1 shows inhibition zone of TiO₂ NPs against pathogens under visible light. Figs. 4–6 show well diffusion assay for TiO₂ NPs against Gram positive *Staphylococcus aureus*, *Streptococcus pneumoniae* and *Bacillus subtilis* bacteria; Gram negative *Proteus vulgaris*, *Pseudomonas aeruginosa* and *E. coli*

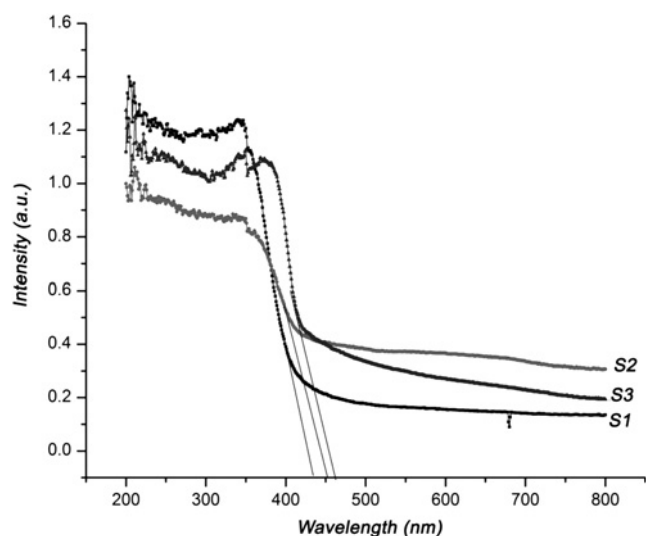


Fig. 3 Absorption spectra for TiO₂ NP samples

Table 1 Zone of inhibition of TiO₂ NPs against pathogens

Microorganism	S1				S2				S3			
	Zone of inhibition, cm											
	Control	Concentrations, µg/ml			Control	Concentrations, µg/ml			Control	Concentrations, µg/ml		
		150	200	250		150	200	250		150	200	250
<i>Streptococcus Pneumonia</i>	3.5	—	2.1	2.3	3.4	—	1.9	2.3	3.4	—	1.8	2.1
<i>Staphylococcus Aureus</i>	3.1	0.6	1.9	2.1	3.2	0.6	1.8	2.2	2.7	—	1.6	2.1
<i>Bacillus subtilis</i>	3.3	—	1.4	1.6	3.2	—	1.6	1.8	3.2	—	1.4	1.8
<i>Proteus Vulgaris</i>	2.1	—	1.5	1.9	1.6	—	1.7	2.0	1.7	—	1.6	1.7
<i>Pseudomonas aeruginosa</i>	2.1	1.1	2.0	2.2	2.4	—	1.8	2.1	2.1	—	2.2	2.4
<i>E. coli</i>	2.3	—	1.6	1.8	2.3	—	1.4	2.0	2.2	—	1.3	1.6
<i>Candida Albicans</i>	3.4	—	2.2	2.5	4	0.6	2.4	2.6	3.2	—	2.0	2.1

^aSensitive – 1.9 cm; Intermediate – 1.5–1.8 cm; Resistance – <1.4 cm.

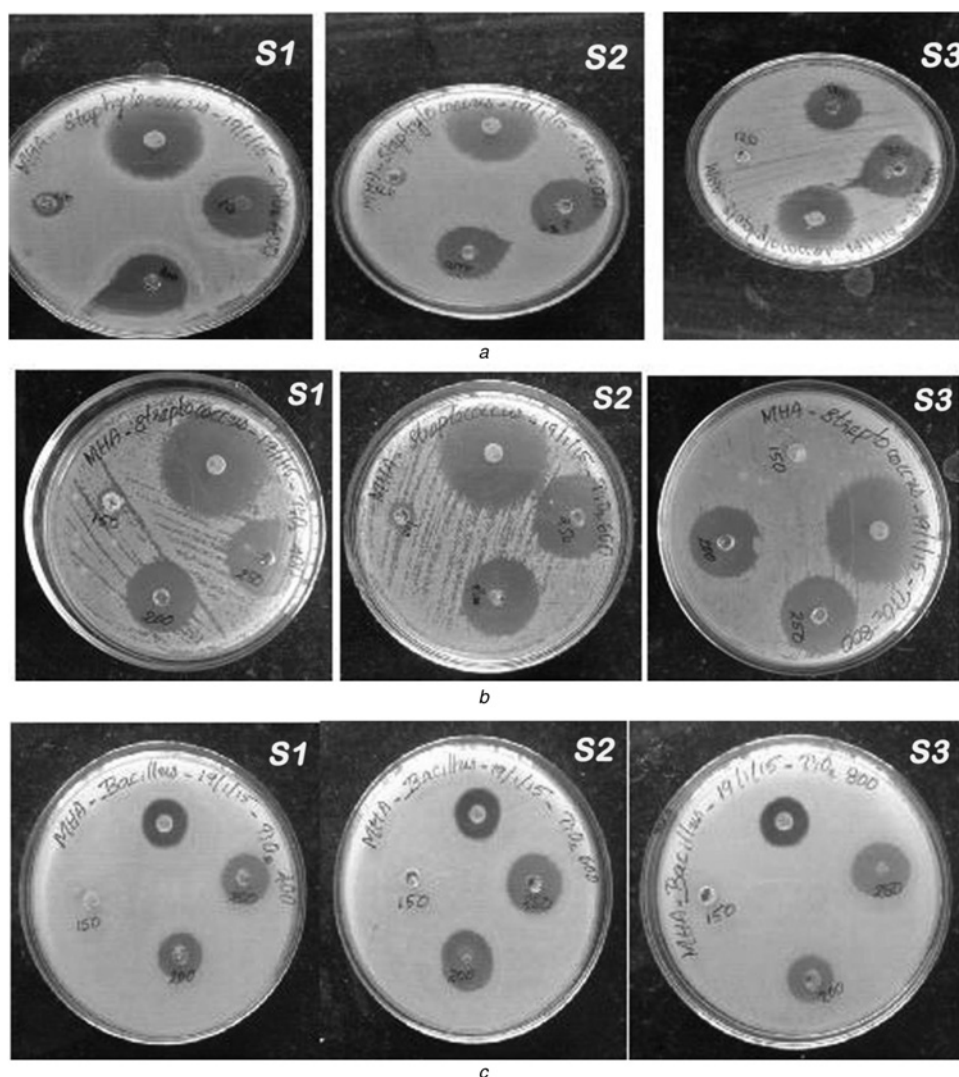


Fig. 4 Well diffusion assay for TiO_2 NPs against

a *Staphylococcus aureus*
b *Streptococcus pneumoniae*
c *Bacillus subtilis*

bacteria and *Candida albicans* fungus respectively in visible light. The sample S1 (anatase) shows sensitivity to *Streptococcus pneumoniae* (2.3 cm), *Staphylococcus aureus* (2.0 cm), *Proteus vulgaris* (1.9 cm) and *Pseudomonas aeruginosa* (2.2 cm) bacteria, and *Candida albicans* (2.5 cm) fungus (Table 1 and Fig. 3–6). The sample S2 (TiO_2 -600) shows sensitivity to *Streptococcus pneumoniae* (2.3 cm), *Staphylococcus aureus* (2.2 cm), *Proteus vulgaris* (2 cm), *Pseudomonas aeruginosa* (2.1 cm) and *E. coli* (2.0 cm) bacteria, and *Candida albicans* (2.6 cm) fungus. However, Sample S3 (rutile) shows sensitivity to *Streptococcus pneumoniae* (2.1 cm), *Staphylococcus aureus* (2.1 cm) and *Pseudomonas aeruginosa* (2.4 cm) bacteria and *Candida albicans* (2.1 cm) fungus only. It can be seen from Table 1 and Figs. 4–6 that the sample S1, (anatase TiO_2), shows significant antimicrobial activity compared with samples S2 and S3 (rutile TiO_2). Similar results are reported in literature [28]. The calcination of TiO_2 NPs causes increase of particle size, which in turn results in the reduction of antimicrobial potency.

The antimicrobial activity of sample S3 is found higher inhibition zone than the standard antibiotics Tetracycline for *Pseudomonas aeruginosa*. All the samples show higher sensitivity to pathogens at 250 μg concentration. No samples found inhibited or sensitive against *Bacillus subtilis* bacteria. TiO_2 NPs samples calcined at various temperatures (400, 600 and 800°C) possesses different sensitivity against microbes due to size variation caused by

calcination process. The present studies found that presence of light, concentration of TiO_2 and size of NPs are the factors for efficient antimicrobial activity against pathogens. Moreover, anatase TiO_2 NPs samples showed considerably good activity against *Streptococcus pneumoniae*, *Staphylococcus aureus*, *Proteus vulgaris* and *Pseudomonas aeruginosa* bacteria and *Candida albicans*.

Antibacterial activity of TiO_2 NPs can be attributed to the decomposition of bacterial outer membranes by reactive oxygen species (ROS), primarily hydroxyl radicals ($\text{OH}^\cdot$), which leads to phospholipid peroxidation and finally cell death [20–23, 29]. Photocatalytic oxidation reactions are initiated when a photon of energy level higher or equal to the band gap energy is absorbed by a TiO_2 catalyst. This reaction promotes an electron (e^-) from the valence band to the conduction band with simultaneous generation of a positive hole (h^+) in the valence band. The positive hole of TiO_2 breaks water molecule to form hydrogen gas (H_2) and hydroxyl radicals ($\text{OH}^\cdot$). The negative-electron reacts with atmospheric oxygen molecule to form super oxide ions. These hydroxyl radicals contact with each other to produce hydroxyl peroxide (H_2O_2). These ROS can decompose organic compounds and extinguish cellular activity. TiO_2 photocatalytic action induces significant damages in cell membranes by ROS, followed by loss of essential functions [25, 26, 28].

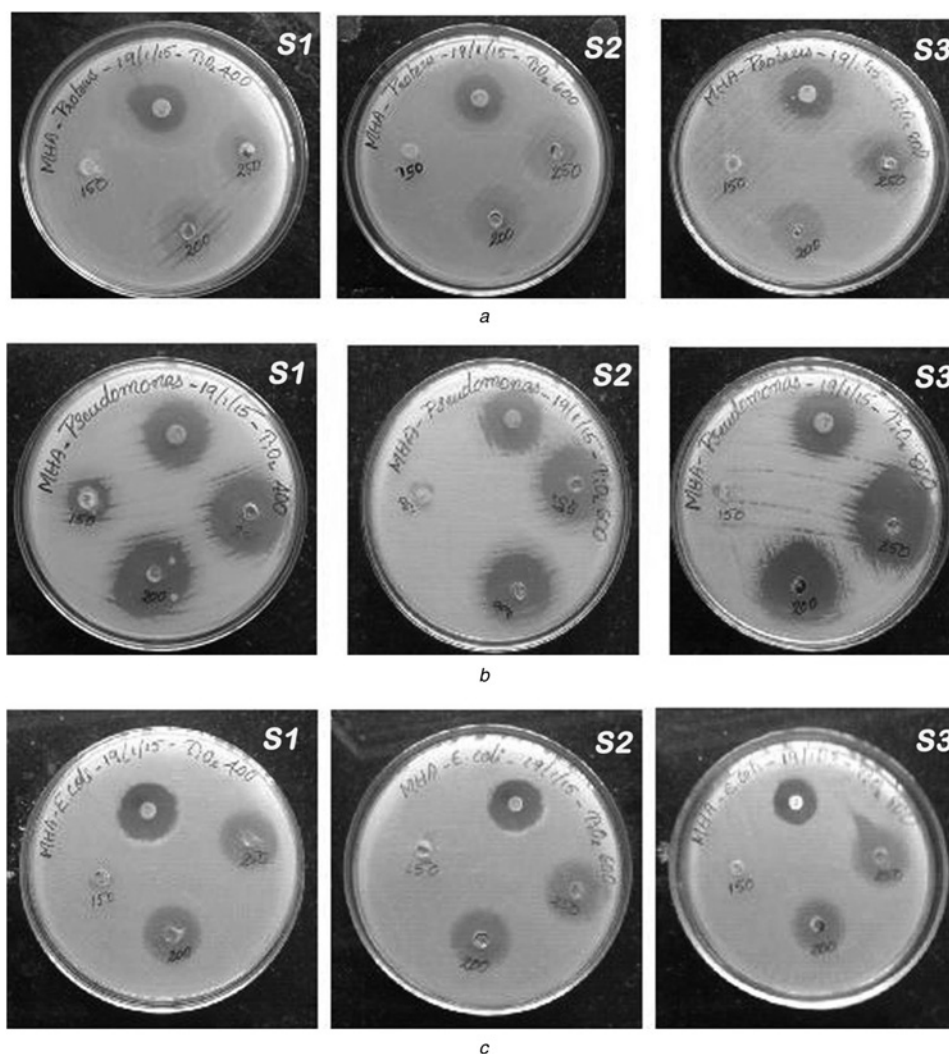


Fig. 5 Well diffusion assay for TiO_2 NPs against

a *Proteus vulgaris*

b *Pseudomonas aeruginosa*

c *Escherichia coli*

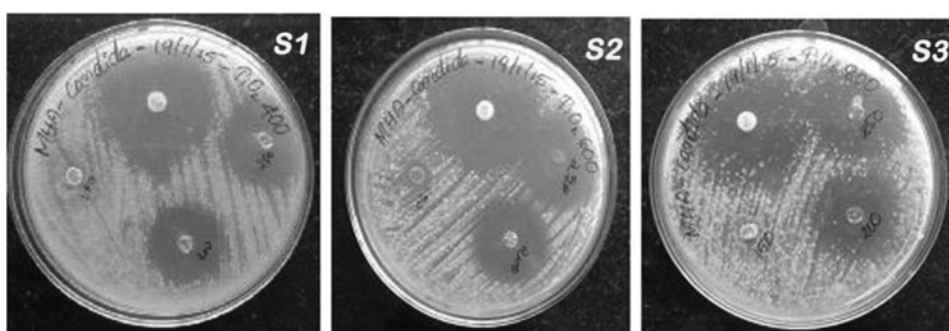


Fig. 6 Well diffusion assay for TiO_2 NPs against *Candida albicans*

The present investigation found that TiO_2 NPs of suitable size may act as promising antibacterial and antifungal agents in future. Further, the present experimental work is done at day light hence, it is important in terms of potential applications of these NPs for antimicrobial modification of various surfaces. In hospital environments, where resistant strains could be transmitted easily and cause infection in patients with surgical wounds and burnt, TiO_2 NPs could be used as the suitable disinfectant. Moreover, it could be possible to manufacture the suture or wound bands by cotton fabrics with antibacterial material TiO_2 to decrease the rate of infection in patients.

4 Conclusions

TiO_2 NPs samples were synthesised by sol-gel method and were characterised by XRD, SEM and UV-Vis spectroscopy. TiO_2 NPs calcined at 400, 600 and 800°C were used for antimicrobial study of Gram positive bacteria (*Staphylococcus aureus*, *Streptococcus pneumonia*, *Bacillus subtilis*), Gram negative bacteria (*Proteus vulgaris*, *Pseudomonas aeruginosa*, *E. coli*) and fungal test pathogen *Candida albicans*. The synthesised TiO_2 NPs were found to have

antimicrobial potency in day light against five bacterial strains *Streptococcus pneumonia*, *Staphylococcus aureus*, *Pseudomonas aerogenosa*, *Proteus vulgaris* and *E. coli* and a fungal strain *Candida albicans*. The calcination of TiO₂ NPs results changes in particle size and optical band gap, which in turn changing antimicrobial potency.

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